Gaussian discrimant analysis

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1 Introduction

Gaussian discriminant analysis (GDA) is a generative classifier. Compared to Naive Bayes, GDA is restricted to continuous features and assumes features are normally distributed within classes, i.e.,

$$p(X|y=c,\theta) = \mathcal{N}(X|\mu_c, \Sigma_c) \tag{1.1}$$

If Σ_c is diagonal, this is equivalent to Naive Bayes. With $k-1+k\left(m+\frac{m(m+1)}{2}\right)$ parameters, GDA has more parameters than Naive Bayes ($k-1+2km$ parameters). Thus, it is more accurate but is the risk of overfitting is higher. In comparison with Naive Bayes,

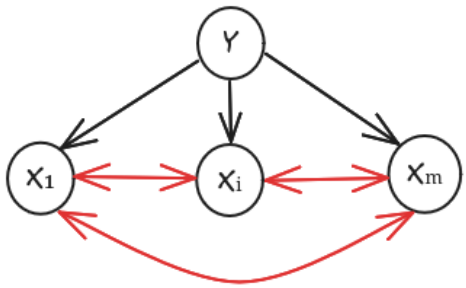

FIGURE 1.1 – Quadratic discriminant analysis with $k-1+k\left(m+\frac{m(m+1)}{2}\right)$ parameters

As,

$$p(Y=c|X,\theta) = \frac{p(Y=c|\theta)p(X|Y=c,\theta)}{p(X|\theta)} \tag{1.2}$$

we can classify a feature vector using the following decision rule :

$$\hat{y}(x) = \arg\max_c \left[\log p(y=c|\pi) + \log p(x|\theta_c)\right] \tag{1.3}$$

2 Multivariate normal

The pdf for an MVN in D dimensions is defined by the following :

$$\mathcal{N}(x|\mu,\Sigma) = \frac{1}{(2\pi)^{D/2}|\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(x-\mu)^T\Sigma^{-1}(x-\mu)\right) \tag{2.1}$$

The expression inside the exponent is the Mahalanobis distance between a data vector x and the mean vector μ. The first two moments are :

$$\mathbb{E}(X) = \mu \quad \text{and} \quad \text{cov}(X) = \mathbb{E}((X-\mu)(X-\mu)^T) \tag{2.2}$$

If we have N i.i.d. samples $x_i \sim (\mu, \Sigma)$, then the MLE for the parameters is given by :

$$\hat{\mu}_{mle} = \frac{1}{N}\sum_{i=1}^{N} x_i = \bar{x} \tag{2.3}$$

$$\hat{\Sigma}_{mle} = \frac{1}{N}\sum_{i=1}^{N}(x_i - x)(x_i - x)^T = \frac{1}{N}\sum_{i=1}^{N} x_i x_i^T - \bar{x}\bar{x}^T \tag{2.4}$$

3 MLE for discriminant analysis

The log-likelihood function is as follows :

$$\log p(\mathcal{D}|\theta) = \left[\sum_{i}^{N}\sum_{c=1}^{C}\mathbb{1}(y_i = c)\log \pi_c\right] + \sum_{c=1}^{C}\left[\sum_{i:y_i=c}\log \mathcal{N}(X|\mu_c, \Sigma_c)\right] \tag{3.1}$$

We see that this factorizes into a term for π, and C terms for each μ_c and Σ_c. Hence we can estimate these parameters separately. For the class prior, we have : $\hat{\pi}_c = \frac{N_c}{N}$ as with Naive Bayes with $N_c = \sum_i \mathbb{1}(y_i = c)$ is the number of examples in class c. For the class-conditional densities, we just partition the data based on its class label, and compute the MLE for each Gaussian :

$$\hat{\mu}_c = \frac{1}{N_c}\sum_{i:y_i=c} x_i = \frac{1}{N_c}\sum_{i=1}^{N}\mathbb{1}(y_i = c)x_i \tag{3.2}$$

$$\hat{\Sigma}_c = \frac{1}{N_c}\sum_{i:y_i=c}(x_i - \hat{\mu}_c)(x_i - \hat{\mu}_c)^T = \frac{1}{N_c}\sum_{i=1}^{N}\mathbb{1}(y_i = c)(x_i - \hat{\mu}_c)(x_i - \hat{\mu}_c)^T \tag{3.3}$$

4 Quadratic discriminant analysis (QDA)

From equation 4.1, we derive :

$$p(Y = c|X, \theta) \propto \pi_c \frac{1}{|\Sigma_c|^{1/2}} \exp\left[-\frac{1}{2}(x - \mu_c)^T \Sigma_c^{-1}(x - \mu_c)\right] \tag{4.1}$$

This formula requires Σ_c to be well conditioned.
This leads to class boundaries between classes c_1 and c_2 being quadratic surfaces :

$$\log(\pi_{c_1}|\Sigma_{c_1}|^{1/2}) - \frac{1}{2}(x - \mu_{c_1})^T {\Sigma_{c_1}}^{-1}(x - \mu_{c_1}) = \log(\pi_{c_2}|\Sigma_{c_2}|^{1/2}) - \frac{1}{2}(x - \mu_{c_2})^T {\Sigma_{c_2}}^{-1}(x - \mu_{c_2}) \tag{4.2}$$

The result is known as **quadratic discrimant analysis (QDA)**

5 Linear discriminant analysis (LDA)

We now consider a special case in which the covariance matrices are **tied** or **shared** across classes (**homoscedasticity**), $\Sigma_c = \Sigma$. In this case, we can simplify equation 1.1 as follows :

$$p(X|y = c, \theta) = \mathcal{N}(X|\mu_c, \Sigma) \tag{5.1}$$

LDA has $k - 1 + km + 1\frac{m(m+1)}{2}$ parameters, is less accurate than QDA but has lower risk of overfitting. The MLE for Σ is :

$$\hat{\Sigma} = \frac{1}{N}\sum_{c=1}^{C}\sum_{i:y_i=c}(x_i - \hat{\mu}_c)(x_i - \hat{\mu}_c)^T \tag{5.2}$$

6 Mahanalobis distance and transformation : another way to interpret LDA

We performe an eigendecomposition of Σ. We write $\Sigma = U\Lambda U^T$ where U is an orthonormal matrix of eigenvectors satisfying $U^T U = I$, and Λ is a diagonal matrix of eigenvalues $\text{Diag}(\sigma_i^2)$. Let ϕ be the **Mahalanobis transformation** (rotation U^T + scaling on new axes) :

$$\phi : x \mapsto z = \Sigma^{-\frac{1}{2}} x = \text{Diag}(\sigma_i^{-1}) U^T x \tag{6.1}$$

The Mahanalobis distance between x_1 and x_2 can be written in two forms :

$$d_M(x_1, x_2) = \sqrt{(x_1 - x_2)^T \Sigma^{-1} (x_1 - x_2)} \tag{6.2}$$

$$= \sqrt{(\phi(x_1) - \phi(x_2))^T (\phi(x_1) - \phi(x_2))} \tag{6.3}$$

Then ϕ decorrelates samples :

$$\mathbf{x} \sim \mathcal{N}(\mathbf{0}, \Sigma) \quad \Rightarrow \quad \phi(x) \sim \mathcal{N}(0, I) \tag{6.4}$$

LDA chooses the closest μ_c in the Mahanalobis space as :

$$p(Y = c|X) \propto \pi_c |\Sigma^{-1}| exp\left[-\frac{1}{2} d_M(X, \mu_c)\right] \tag{6.5}$$

We show below that LDA class boundaries are hyperplanes. Indeed, using z, Mahanalobis transformation and inspiring from equation 4.2, we have :

$$z \in \text{boundary between class 1 and 2} \tag{6.6}$$

$$\Leftrightarrow \frac{1}{2}\|z - z_1\|^2 - \log \pi_1 = \frac{1}{2}\|z - z_2\|^2 - \log \pi_2 \tag{6.7}$$

$$\Leftrightarrow \left(z - \frac{z_1 + z_2}{2}\right)(z_2 - z_1) = \log \frac{\pi_1}{\pi_2} \tag{6.8}$$

7 LDA as a dimensionality reduction method

TODO : slide 10

8 Strategies for preventing overfitting

The speed and simplicity of the MLE method is one of its greatest appeals. However, the MLE can badly overfit in high dimensions. In particular, the MLE for a full covariance matrix is singular if $N_c < m$. And even when $N_c > m$, the MLE can be ill-conditioned, meaning it is close to singular. There are several possible solutions to this problem :

- Use a diagonal covariance matrix for each class, which assumes the features are conditionally independent ; this is equivalent to using a naive Bayes classifier
- Use a full covariance matrix, but force it to be the same for all classes, $\Sigma_c = \Sigma$. This is an example of parameter tying or parameter sharing, and is equivalent to LDA

- Use a diagonal covariance matrix and forced it to be shared. This is called diagonal covariance LDA
- Use a full covariance matrix, but impose a prior and then integrate it out. If we use a conjugate prior, this can be done in closed form ; this is analogous to the Bayesian Naive Bayes
- Fit a full or diagonal covariance matrix by MAP estimation.
- Project the data into a low dimensional subspace and fit the Gaussians there.

9 Regularized LDA

TODO

10 Diagonal LDA

Covariance matrices are tied, so $\Sigma_c = \Sigma$ as in LDA, and then to use a diagonal covariance matrix for each class. The hypothesis can be view as homoscedasticity (LDA) and conditional independence (Naive Bayes). Diagonal LDA has $k - 1 + km + m$ parameters. Thus, we have even less parameters than Naive Bayes. Diagonal LDA is very simplistic but very robust.

11 Shrinkage and high dimensional statistics

TODO

12 References

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